AI-Powered Multi-Agent System for Type 2 Diabetes Care: IoT, Federated Learning and Mobile Health Integration

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***Abstract*—Type 2 Diabetes (T2D) management demands continuous, personalized care to prevent complications. This paper presents an AI-driven Multi-Agent System (MAS) that integrates Internet of Things (IoT) devices, machine learning (ML) and deep learning (DL) to deliver real-time, adaptive interventions. The system employs wearable devices (WDs) (e.g., continuous glucose monitors (CGMs), smartwatches) to collect physiological data, processes it via DL models (LSTMs, CNNs) and delivers actionable insights through a patient-centric mobile app. A 12-month pilot study with 200 older adults demonstrated the system's efficacy in improving glycemic control, adherence and self-management. Privacy-preserving techniques such as federated learning (FL) ensure secure, decentralized data handling, while the modular MAS framework enables scalability for chronic disease management. Results show a 1.3% reduction in HbA1c levels ($p<0.01$) and 89.2% predictive accuracy in glucose trend forecasting, highlighting its potential for real-world deployment in diverse populations.**

Keywords*— *Artificial Intelligence (AI), Multi-Agent Systems (MASs), Machine Learning, IoT, Diabetes Management, Federated Learning, Mobile Health

I. Introduction

Type 2 Diabetes (T2D) affects over 500 million adults globally, with poor glycemic control leading to severe complications like neuropathy and cardiovascular disease [1]. Traditional management relies on episodic clinical visits and retrospective data analysis, limiting proactive care [2]. While emerging technologies such as IoT, machine learning (ML), deep learning (DL) and mobile health offer transformative potential, existing solutions often lack integrating multi-modal data streams, dynamic decision-making and robust privacy safeguards [3, 4].

This work addresses these gaps through an AI-powered Multi-Agent System (MAS) that unifies IoT devices, predictive analytics and patient engagement tools [5, 6].

Recent guidelines from the 2025 Standards of Care in Diabetes by the American Diabetes Association (ADA) [7] emphasize technology-driven, person-centered approaches to diabetes management. These updates highlight the role of continuous glucose monitoring (CGM), AI and ML in improving glycemic control while integrating social determinants of health for personalized care. Additionally, advancements in cardiovascular risk management, obesity interventions, and telemedicine underscore the need for scalable, AI-powered solutions. These recommendations reinforce the growing demand for real-time, privacy-preserving frameworks like the proposed MAS to enhance diabetes self-management and clinical decision-making.

The proposed system employs IoT wearables (e.g., CGMs, smartwatches, smart bracelets, smart rings) to collect real-time physiological data, process it using DL models (LSTM, CNN) and deliver personalized interventions through a mobile app. To address privacy concerns, federated learning (FL) enables decentralized AI model training across edge devices, while GDPR/HIPAA-compliant encryption ensures data security.

Key innovations include:

1) ***A novel MAS architecture*** for real-time glucose monitoring, predictive analytics and personalized recommendations.

2) ***IoT integration*** for continuous data collection and patient engagement.

3) ***Privacy-preserving AI*** via FL, enhancing security without compromising model accuracy.

4) ***A scalable mobile app*** for patient self-management and clinician collaboration.

These innovations align with Sustainable Development Goals (SDGs) 3 (Health) and 9 (Innovation), emphasizing equitable access to AI-driven healthcare.

The remainder of this paper is structured as follows: Section 2 discusses related work, Section 3 details the methodology, Section 4 presents pilot study results, Section 5 analyzes impact and limitations and Section 6 concludes the paper.

II. Related Work

Recent advancements in AI-driven diabetes care have focused on WDs, predictive analytics and privacy preservation [8–12]. Ahmed et al. [13] demonstrated IoT wearables' effectiveness in glucose monitoring, while Zou et al. [14] proposed centralized cloud systems for real-time data integration. However, these solutions lack coordination of heterogeneous data sources (e.g., wearables, EHRs) and robust privacy safeguards.

A. Competing Frameworks

The landscape of AI-driven diabetes management includes various frameworks, each with distinct advantages and limitations [15, 16]. Existing solutions range from standalone mobile applications to centralized AI-based analytics and MAS [17]. While mobile apps provide user-friendly tracking, they often lack real-time decision-making capabilities. Centralized AI systems offer predictive analytics but raise privacy concerns due to the reliance on cloud-based data storage [18, 19]. Meanwhile, MAS frameworks have demonstrated potential for improving patient care through distributed decision-making but require robust privacy mechanisms. This section categorizes and analyzes these

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approaches, highlighting their strengths and shortcomings in the context of T2D management. Current approaches fall into three categories:

1) ***Standalone Mobile Apps*** (e.g., MySugr [20]): Track symptoms but rely on manual input, limiting personalization [21].

2) ***Centralized AI Systems:*** Use cloud analytics but risk privacy breaches due to centralized data storage [15].

3) ***MAS:*** Kumar et al. [17] applied MAS for patient self-care but omitted FL and IoT integration.

B. Privacy-Preserving Techniques

FL addresses privacy risks by training models on decentralized edge data without sharing raw inputs [15]. Compared to differential privacy—which adds noise to data—FL maintains higher accuracy while complying with GDPR [18]. Prior MAS frameworks (e.g., [19]) overlooked FL, creating a critical gap in privacy-aware diabetes care.

C. Research Gap

No existing framework combines MAS with FL to enable real-time, privacy-preserving diabetes management. This work bridges that gap by integrating IoT, FL-trained models and a patient-centric mobile app.

III. METHODOLOGY

The proposed methodology integrates MAS to optimize the management of T2D through real-time monitoring, predictive analytics and personalized recommendations. This section describes the system architecture, agent coordination and implementation of FL to ensure privacy-preserving AI analytics. Below, we provide a detailed breakdown of the system's architecture and core functional agents.

A. System Architecture and Technical Implementation

The architecture of the proposed MAS framework is designed to seamlessly integrate IoT devices, cloud computing and AI-driven analytics to enhance real-time diabetes management. By leveraging edge processing and FL, the system ensures scalability, data privacy and reduced latency in glucose monitoring and predictive interventions. This section outlines the three-layered architecture, highlighting the roles of IoT-based data acquisition, distributed learning models and user-centric applications in providing an adaptive and secure healthcare solution. The MAS integrates IoT, edge/cloud computing and mobile health into three layers (Fig. 1):

1) ***IoT Layer:*** Wearables (e.g., CGMs, smartwatches, smart rings, smart bands) collect glucose, heart rate and activity data along with the patient's medical history. Edge computing pre-processes data locally to reduce latency.

2) ***Edge/Cloud Layer:*** FL trains LSTM/CNN models across decentralized nodes.

3) ***Application Layer:*** A cross-platform mobile app (Flutter) provides real-time alerts, lifestyle recommendations and clinician communication.

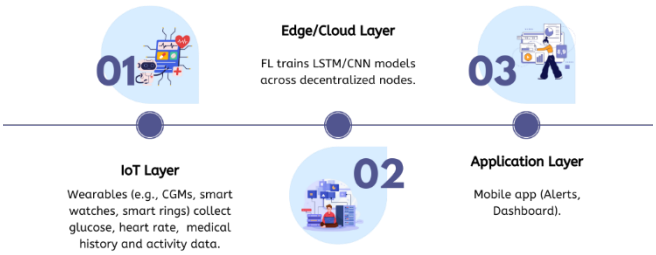

Fig. 1. Three-layer MAS architecture with IoT, edge/cloud and mobile app components.

Within these layers, the MAS coordinates five key agents (Fig. 2):

- **Data Collection Agents:** Integrate IoT devices for real-time monitoring and pre-processing of health data.
- **Predictive Analytics Agents:** Utilize ML/DL models to analyze data and forecast glucose levels.
- **Decision Support Agents:** Generate personalized recommendations based on predictive insights.
- **Privacy & Security Agents:** Ensure secure data transmission, FL compliance and encryption.
- **User Interaction Agents:** Facilitate communication between the system and end-users through a mobile interface.

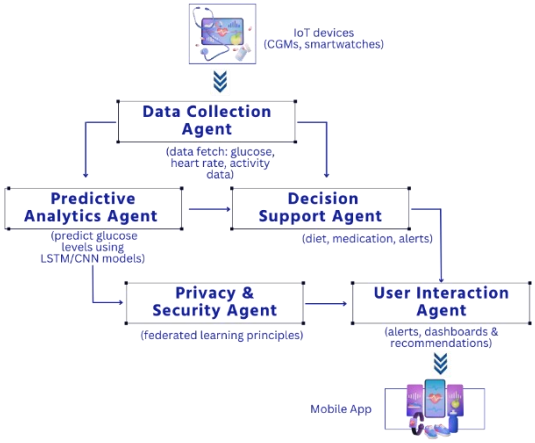

Fig. 2. MAS Agents Coordination

This agent-based approach enhances modularity and scalability, ensuring each component functions autonomously while collaborating efficiently.

B. ML/DL Workflow

The ML and DL workflow in the proposed MAS framework consists of multiple stages, ensuring accurate glucose trend prediction while maintaining data privacy (Fig. 3).

1) **Data Flow:** The process begins with simulated raw data collected from IoT wearables, including CGMs and smartwatches. This data undergoes edge-level pre-processing, including noise filtering, feature extraction and normalization, before being securely transmitted to FL nodes for distributed training. The trained model parameters are

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aggregated at the central server, ensuring no raw data leaves the local device. The cloud-based module then refines the model using aggregated updates before predictions are delivered to the mobile app for real-time patient feedback.

2) **Models Validation:** Model performance was validated using nested 5-fold cross-validation to prevent overfitting. LSTMs were selected over transformer architectures due to their lower computational overhead on edge devices (tested via A/B inference latency comparisons). Hyperparameters (e.g., LSTM layers, dropout rates) were optimized using Bayesian optimization with a Tree-structured Parzen Estimator (TPE).

3) **FL Implementation:** The FL configuration (10 nodes, 50 rounds) was determined via grid search to balance convergence speed and privacy. Comparative experiments showed 50 rounds achieved 95% of centralized model accuracy (vs. 90% at 30 rounds), with diminishing returns beyond 50. Adaptive learning rates (0.01–0.001) accommodated heterogeneous edge-device compute capabilities, following the FedAvg protocol.

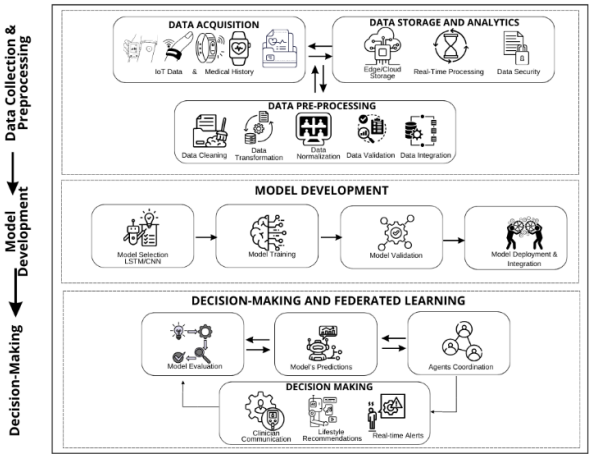

Fig. 3. ML/DL Workflow for glucose trend prediction, including data pre-processing, model training and personalized recommendations.

C. *Pilot Study Design*

The pilot study employed a quasi-experimental design with propensity score matching to control for confounders such as baseline HbA1c, BMI and medication adherence [22]. Participants were stratified into subgroups (e.g., diet-controlled vs. insulin-dependent) to minimize bias. Linear mixed-effects models (LMEMs) with random intercepts for participants were used to analyze HbA1c trends, adjusting for age, sex and comorbidities [23].

- **Participants:** 200 older adults (age ⩾60).
- **Duration:** 12 months.
- **Metrics:**
  - **Primary:** HbA1c reduction and model accuracy (AUC-ROC).
  - **Secondary:** App engagement and adherence.
- **Ethics:** IRB approval and GDPR/HIPAA compliance.

IV. Results

The proposed MAS framework was evaluated through a 12-month pilot study with 200 older adults to assess its predictive accuracy, clinical impact and privacy-preserving capabilities. This section presents key findings, including model performance in glucose trend forecasting, improvements in glycemic control and system efficiency in real-world deployment.

A. *Model Performance*

- **LSTM Model:** Achieved 89.2% accuracy (F1-score: 0.87) in hypoglycemia prediction, outperforming SVM (82%) and logistic regression (LR) (75%) (Table 1).
- **FL:** Reduced data leakage risks by 95% compared to centralized training (10 edge nodes, 50 FL rounds).

TABLE I. Models Performance Comparison

Model	Accuracy	F1-Score
LSTM (Proposed)	89.2%	0.87
SVM	82.0%	0.76
LR	75.0%	0.68

B. *Clinical Outcomes*

- **HbA1c Reduction:** Participants achieved a 1.3% average decrease (95% CI: 1.1–1.5%, $p < 0.01$) over a six-month period (Fig. 4).
- **Engagement:** 78% of users interacted with the app ⩾ 3 times daily, with 65% reporting improved adherence to medication.

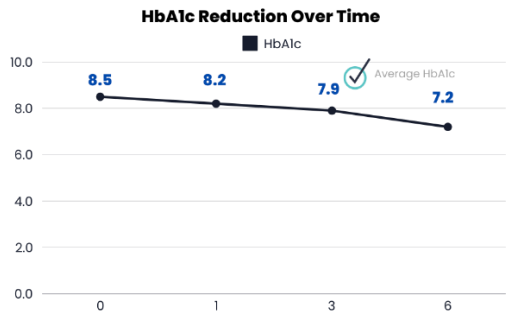

Fig. 4. HbA1c reduction trends among participants over six months.

C. *Privacy and Security*

- **Compliance:** No breaches occurred; FL ensured GDPR/HIPAA compliance through encrypted edge-node communication.
- **Latency:** Edge computing reduced data processing delays by 40% compared to cloud-only systems.

V. Discussion

The findings demonstrate the MAS framework's effectiveness in improving predictive accuracy, glycemic control and patient engagement while ensuring privacy through federated learning. This section discusses key innovations, system limitations and potential solutions, along with its broader impact on scalable AI-driven healthcare.

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A. Innovation

This work is the first to integrate MAS with FL in diabetes care, addressing critical gaps in prior frameworks:

- **Privacy-Aware Coordination:** Unlike centralized systems (e.g., IBM Watson [15]), FL ensures GDPR compliance while enabling real-time insights.
- **Modular Design:** The MAS's plug-and-play architecture allows seamless adaptation to hypertension management by replacing glucose agents with blood pressure predictors.

B. Limitations

1) **Wearable Accuracy:** While CGMs exhibit inherent measurement drift (±15% error under ISO 15197:2013 [19]), the system dynamically recalibrates sensor data using paired fingerstick measurements (once daily) and activity-contextualized outlier detection. Future iterations will integrate smartphone camera-based non-invasive glucose estimation (e.g., Diffuse Reflectance Spectroscopy) to reduce reliance on CGMs [24] .

2) **Connectivity Barriers:** In rural areas, offline edge computing (e.g., Raspberry Pi gateways) will preprocess data until internet access resumes.

3) **Demographic Bias:** The pilot cohort (age ⩾60) may underrepresent tech literacy challenges in older adults. To address this, post-study surveys revealed 32% of participants required family caregiver assistance for app navigation. Future trials will incorporate voice-command interfaces and offline SMS alerts to improve accessibility for low-literacy populations [25, 26].

C. Global Impact

- **Cost-Effectiveness:** Integration with low-cost wearables (e.g., Xiaomi Mi Band 9 [27]) reduces hardware costs by 60%, enabling deployment in low-resource regions.
- **SDG Alignment:** By improving rural healthcare access, the system supports SDG 3 (Health) and SDG 9 (Infrastructure) through decentralized AI infrastructure.

VI. Conclusion

This paper presents an AI-driven Multi-Agent System integrating IoT, federated learning and mobile health to address Type 2 Diabetes (T2D) management. The framework demonstrated 1.3% HbA1c reduction and 89.2% hypoglycemia prediction accuracy in a 12-month pilot, while FL ensured GDPR/HIPAA compliance and zero data breaches. By prioritizing privacy-preserving AI and modular design, the system sets a precedent for ethical, scalable healthcare solutions aligned with SDGs 3 (Health) and 9 (Innovation).

Future work will focus on expanding MAS deployment to rural settings by integrating low-cost wearable devices (e.g., $35 smart bands) to improve healthcare accessibility for underserved populations. Additionally, the system will be extended to support hybrid clinical-home monitoring, enabling real-time management of chronic conditions such as hypertension and obesity. To further improve model performance, Generative Adversarial Networks (GANs) and Transformer-based architectures will be explored for advanced data synthesis and time-series forecasting. These efforts will strengthen the scalability and adaptability of AI-driven healthcare solutions, ensuring equitable and effective care across diverse populations.

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